

1. Administrative & Study Identification

Full Study Title: A Study of the Effectiveness and Safety of Nivolumab Plus Chemotherapy in Participants With Untreated Advanced/Recurrent Gastric Cancer
Short Title/Acronym: G-KNIGHT
Protocol Number: UMIN000046820
NCT Number: NCT05334719
Sponsor Name: Bristol-Myers Squibb (in collaboration with Ono Pharmaceutical Co. Ltd.)
Investigational Product: Nivolumab plus Chemotherapy (SOX, CapeOX, or FOLFOX)
Phase of Development: Observational / Real-World Study
Medical Monitor: Bristol-Myers Squibb Study Director

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the effectiveness, safety, and actual treatment status of first-line nivolumab plus chemotherapy in participants with untreated advanced/recurrent gastric cancer in a Japanese real-world setting. **Secondary Objectives:** To evaluate Overall Survival (OS), Duration of Response (DOR), Duration of Treatment (DOT), Time to Next Treatment (TNT), and overall incidence of Adverse Events (AEs).

2.2 Background & Rationale Nivolumab plus chemotherapy has shown proven efficacy in clinical trials (e.g., CheckMate 649 and ATTRACTION-4) for human epidermal growth factor receptor 2 (HER2)-negative, treatment-naïve advanced or recurrent gastric cancer (GC). However, real-world utilization data remain limited. This study generates large-scale, real-world evidence to support the effectiveness and safety of this combination in untreated Japanese patients.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Observational (Retrospective and Prospective) **Control Type:** Single-arm (Cohort 1) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N\$:** 527 participants
Number of Sites: 23 Japanese sites **Estimated Study Start:** November 2021 **Estimated Primary Completion:** December 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 20 years. 2. Histologically confirmed advanced or recurrent gastric cancer (GC). 3. Have initiated or are confirmed to receive nivolumab plus chemotherapy (SOX, CapeOX, or FOLFOX) as the first-line therapy between the date of approval of nivolumab plus chemotherapy and December 31, 2022.

4.2 Exclusion Criteria 1. Treated with antineoplastic agents as first-line treatment for advanced or recurrent gastric cancer prior to the initiation of nivolumab plus chemotherapy (participants who have received prior perioperative chemotherapy or bisphosphonates for osseous metastases may be enrolled). 2. Confirmed to be positive for HER2. 3. Diagnosed with gastric cancer and have previously received investigational drugs with anti-tumor effects.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Nivolumab administered in combination with chemotherapy regimens (S-1/oxaliplatin [SOX], capecitabine/oxaliplatin [CapeOX], or 5-fluorouracil/leucovorin/oxaliplatin [FOLFOX]). **Route of Administration:** Intravenous / Oral (depending on the specific chemotherapy agents utilized). **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or patient request. Evaluated for up to approximately 4 years.

5.2 Concomitant & Prohibited Therapy Permitted: Prior perioperative chemotherapy and prior bisphosphonates for osseous metastases. **Prohibited:** Prior first-line antineoplastic agents for advanced or recurrent gastric cancer, or prior investigational drugs with anti-tumor effects.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
:---	:---	Screening Prior to 1st Dose Informed Consent, Histological Confirmation, HER2 Status Check
Baseline	Day 0	Initiation of first-line Nivolumab + Chemotherapy
Interim	Real-world routine	Routine clinical follow-ups (Safety Labs, Response Assessment per SoC)
End of Study	Up to 4 Years	Data cutoff / Final follow-up for Survival and Progression Events

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-free survival (PFS), Objective response rate (ORR), Incidence of all immune-related

adverse events (irAE) according to CTCAE v5.0, and Incidence of treatment-related adverse events (TRAE) leading to treatment discontinuation. **Measurement Method:** Real-world observational data abstraction, assessed for up to approximately 4 years.

7.2 Secondary & Exploratory Endpoints * Overall survival (OS). * Duration of response (DOR) and Duration of treatment (DOT). * Time to next treatment (TNT) and Number of participants receiving subsequent therapy.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard real-world collection of all adverse events (AEs), focusing particularly on immune-related adverse events (irAEs) and treatment-related adverse events (TRAEs). **Serious Adverse Events (SAEs):** Evaluated per standard institutional pharmacovigilance and real-world data collection procedures. **Data Monitoring Committee (DMC):** Quality control (QC) review and data curation managed by the sponsor and respective CROs.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All enrolled patients who met the eligibility criteria and initiated nivolumab + chemotherapy (Real-World Cohort). **Sample Size Justification:** \$N=527\$, based on the observational capture of patients initiating therapy across 23 Japanese sites during the predefined enrollment window. **Handling of Missing Data:** Standard observational research methodologies; subgroup analyses conducted for patients stratified by factors including age and programmed cell death ligand 1 (PD-L1) combined positive score (CPS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with observational research guidelines and the Declaration of Helsinki, registered under UMIN000046820 and ClinicalTrials.gov. **Monitoring Plan:** Retrospective and prospective abstraction of medical records across the 23 participating Japanese sites. **Audit Trail:** Electronic data capture from real-world medical records subject to sponsor QC.

11. Study Identifiers & Governance

Primary ID: UMIN000046820 **Registry Number:** NCT05334719 **Sponsor/Lead Organization:** Bristol-Myers Squibb / Ono Pharmaceutical Co. Ltd. **Study Title:** G-KNIGHT **Official Regulatory Title:** A Study of

12. Clinical Framework (Gastric Cancer Specific)

Primary Condition: Advanced or Recurrent Gastric Cancer (GC)
Diagnosis Threshold: Histologically confirmed, HER2-negative
Medical Methodology: First-line Immunotherapy + Chemotherapy
Combination Optimization Target: Progression-Free Survival (PFS)
and Objective Response Rate (ORR)

13. Study Procedures & Methodology

Management Pathway: Standard real-world clinical pathway for
advanced gastric cancer. **Education Protocol (HCP):** Standard of
care administration protocols for Nivolumab + SOX/CapeOX/FOLFOX.
Education Protocol (Patient): Routine clinical consent and
oncology treatment education. **Quality Audit Mechanism:**
Retrospective/prospective clinical data curation and QC review by
the sponsor.

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 4 years **Onsite Visit Cadence:**
Managed per routine clinical oncology practice for chemotherapy/
immunotherapy monitoring. **Remote Monitoring Frequency:** N/A
(Registry data abstraction) **Checklist Review Interval:** Ongoing
real-world data collection through December 2025.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				